

Low Level Design (LLD)

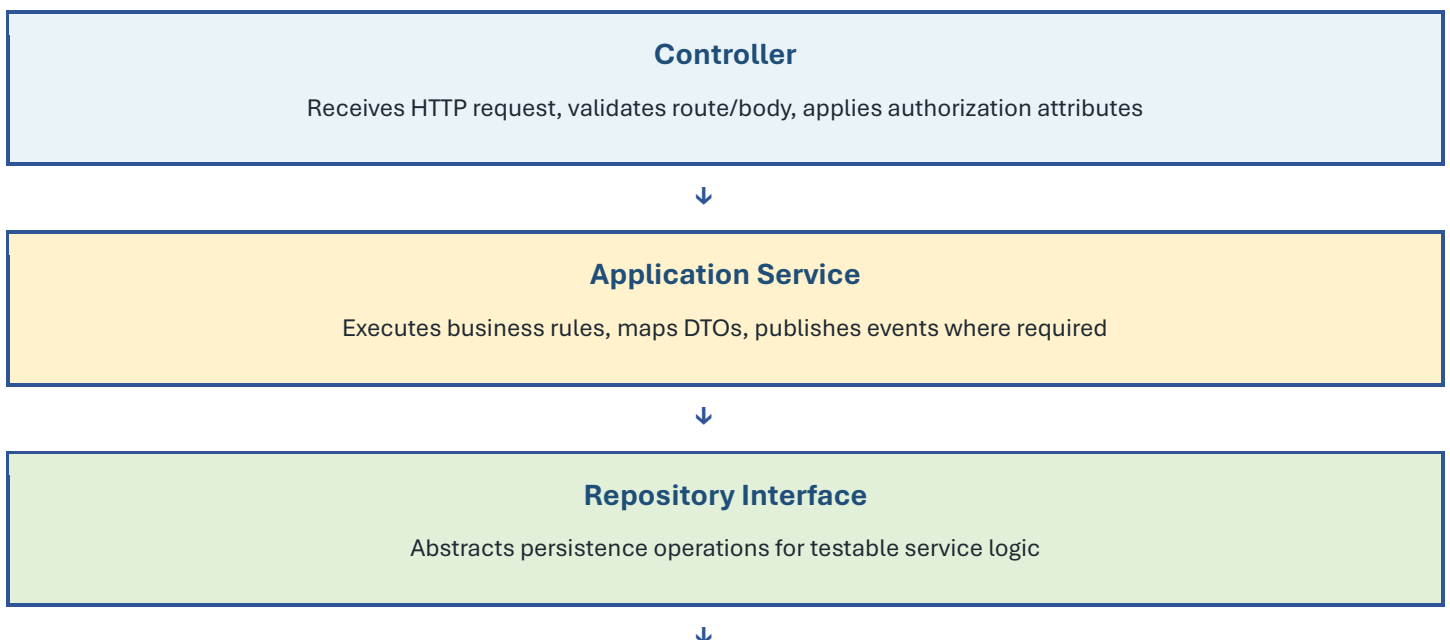
E-Commerce Product Management System

This document explains the internal design of modules, classes, API mappings, data objects, and key execution flows.

1. LLD Scope

Area	Included in LLD
Frontend	Angular feature modules, guards, interceptors, services and route-level responsibilities.
Backend	Controller, service, repository and DbContext-level structure for each API.
Data Model	Important entities and their ownership.
API Mapping	Main endpoint groups and handler responsibilities.
Flows	Login, product create/update, workflow approval, reporting and audit execution flow.
Testing	NUnit low-level validation structure across service test projects.

2. Standard Service Internal Pattern



Repository Implementation

Uses EF Core DbContext for database operations



SQL Server Table

Stores service-owned data only

3. Frontend Low-Level Design

Frontend Layer	Files / Components	Responsibility
Routing	app.routes.ts	Defines public, authenticated and role-protected routes.
App Configuration	app.config.ts	Registers router, HTTP interceptors, toastr and global error handler.
Guards	auth.guard.ts, role.guard.ts	Protect routes based on login status and user role.
Interceptors	auth, token-refresh, error, loading	Attach JWT token, refresh expired tokens, centralize errors and loading state.
Core Services	auth.service, user.service, token-storage.service	Handle API calls, session state and local token storage.
Catalog Feature	product-list, product-form, product-media, product-variants, category-list	Manages catalog administration workflows.
Workflow Feature	product-workflow, approvals, timeline, status badge	Handles submission, approval status and admin approval views.
Reporting/Audit	reporting-dashboard, audit-trail	Displays dashboard reports, exports and audit records.

4. Identity.API LLD

Controller / Class	Collaborators	Responsibility
AuthController	IAuthService	Register, login, refresh, logout, forgot/reset password, change password and profile.
UsersController	IUserService, IPublishEndpoint	Admin user listing, profile update, role change, account status and delete user.
AuthService	IAuthRepository, IUserRepository, JwtTokenGenerator	Authentication business rules, token issuing, refresh token rotation, password reset.
UserService	IUserRepository	User profile, role, active status and user statistics operations.
AuthRepository	IdentityDbContext	User lookup, refresh token and password reset token persistence.
UserRepository	IdentityDbContext	User CRUD and query operations.
JwtTokenGenerator	JwtSettings	Creates signed JWT access tokens and refresh tokens.
PasswordHasher	Static utility	Hashes and verifies user passwords.
Entity	Important Fields	Purpose
User	Id, FullName, Email, PasswordHash, Role, IsActive	Represents application user and role identity.
RefreshToken	Token, UserId, ExpiresAt, IsRevoked	Supports secure session continuation.
PasswordResetToken	Token, UserId, ExpiresAt, IsUsed	Supports forgot/reset password workflow.

5. Catalog.API LLD

Controller / Class	Collaborators	Responsibility
ProductsController	IProductService, IPublishEndpoint	Product create, read, update, delete and audit publishing.
CategoriesController	ICategoryService	Category create, read, update and delete.
ProductVariantsController	IProductVariantService	Variant CRUD for product color, size and barcode.
MediaAssetsController	IMediaAssetService	Product media management.
ProductService	IProductRepository, ICategoryRepository, ISkuGenerator	Product business rules, SKU generation, catalog visibility and reporting events.
SkuGenerator	IProductRepository	Generates unique SKUs from brand/product information.
CatalogDbContext	EF Core	Owns Products, Categories, ProductVariants and MediaAssets tables.
Entity	Important Fields	Purpose
Product	Id, Name, Brand, SKU, Description, PublishStatus, CategoryId	Main catalog item.
Category	Id, Name, Description	Groups products for browsing and reporting.
ProductVariant	Id, ProductId, Color, Size, Barcode	Defines sellable variations of a product.
MediaAsset	Id, ProductId, Url, AltText, SortOrder	Stores product media references.

6. Workflow.API LLD

Controller / Class	Collaborators	Responsibility
WorkflowController	IWorkflowService	Pricing, inventory, submit for review, status update and pending approvals endpoints.
WorkflowService	IWorkflowRepository, IPublishEndpoint	Business rules for pricing, inventory and approval status changes.
WorkflowRepository	WorkflowDbContext	Persists and retrieves Price, Inventory and Approval records.
WorkflowDbContext	EF Core	Applies timestamps and configures workflow entities.
Entity	Important Fields	Purpose
Price	ProductId, MRP, SalePrice, GSTPercent, EffectiveFrom	Stores pricing details for a product.
Inventory	ProductId, WarehouseLocation, AvailableQty, ReservedQty, ReorderLevel	Stores stock and warehouse information.
Approval	ProductId, Status, SubmittedByUserId, ApprovedByUserId, Remarks	Tracks product approval workflow.

7. Reporting.API LLD

Controller / Consumer / Class	Collaborators	Responsibility
ReportsController	IReportingService	Dashboard KPI, product reports, CSV export and snapshots.
AuditController	IAuditService	Audit trail listing by page, product or user.
ReportingService	IReportingRepository, IMemoryCache	Fetches reports, exports CSV and caches dashboard snapshot for 10 minutes.
AuditService	IAuditRepository	Retrieves audit logs.

ProductReportChangedConsumer	IReportingRepository	Synchronizes product report read model from catalog events.
ProductStatusChangedConsumer	IReportingRepository / audit persistence	Handles workflow status change reporting.
AuditLogCreatedConsumer	IAuditRepository	Persists audit log event data.
UserCountChangedConsumer	IReportingRepository	Updates dashboard user count metrics.
Entity	Important Fields	Purpose
ProductReport	ProductId, ProductName, SKU, Status, CategoryName, CreatedByUserId	Read model for reporting product data.
DashboardSnapshot	SnapshotDate, TotalProducts, PublishedProducts, PendingApprovals, RejectedProducts, TotalUsers	Stores KPI snapshots.
AuditLog	EntityName, EntityId, Action, ByUserId, OldValues, NewValues	Stores audit trail with before/after JSON payloads.

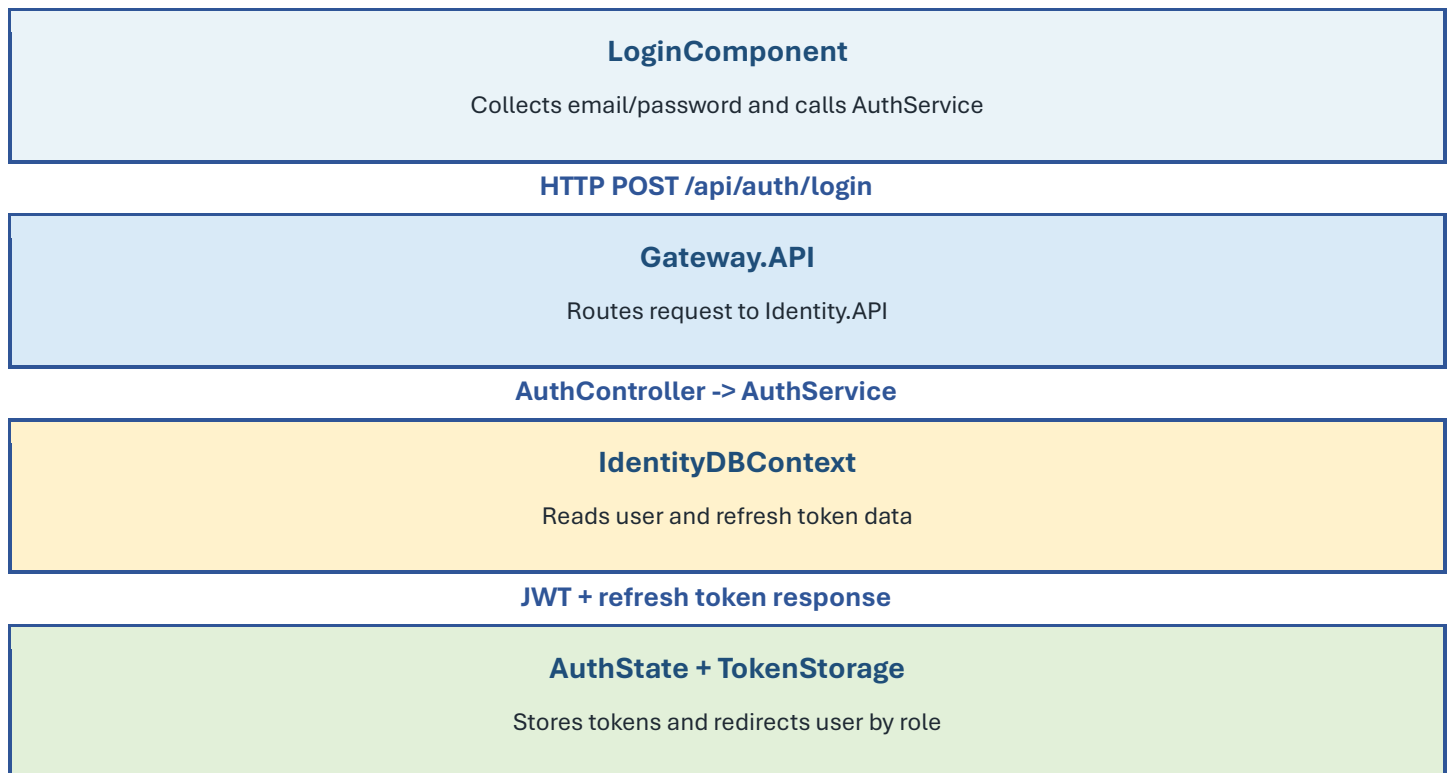
8. API Endpoint Mapping

Endpoint Group	Sample Methods	Controller	Service Method Type
/api/auth	POST login/register/refresh/logout/reset-password	AuthController	Authentication and token methods in AuthService.
/api/users	GET users, PUT role/status, DELETE user	UsersController	User management methods in UserService.
/api/products	GET, POST, PUT, DELETE products	ProductsController	Product CRUD and reporting event methods in ProductService.
/api/categories	GET, POST, PUT, DELETE categories	CategoriesController	Category CRUD methods in CategoryService.
/api/workflow/products/{id}/pricing	GET, PUT pricing	WorkflowController	GetPricingAsync, UpdatePricingAsync.
/api/workflow/products/{id}/inventory	GET, PUT inventory	WorkflowController	GetInventoryAsync, UpdateInventoryAsync.

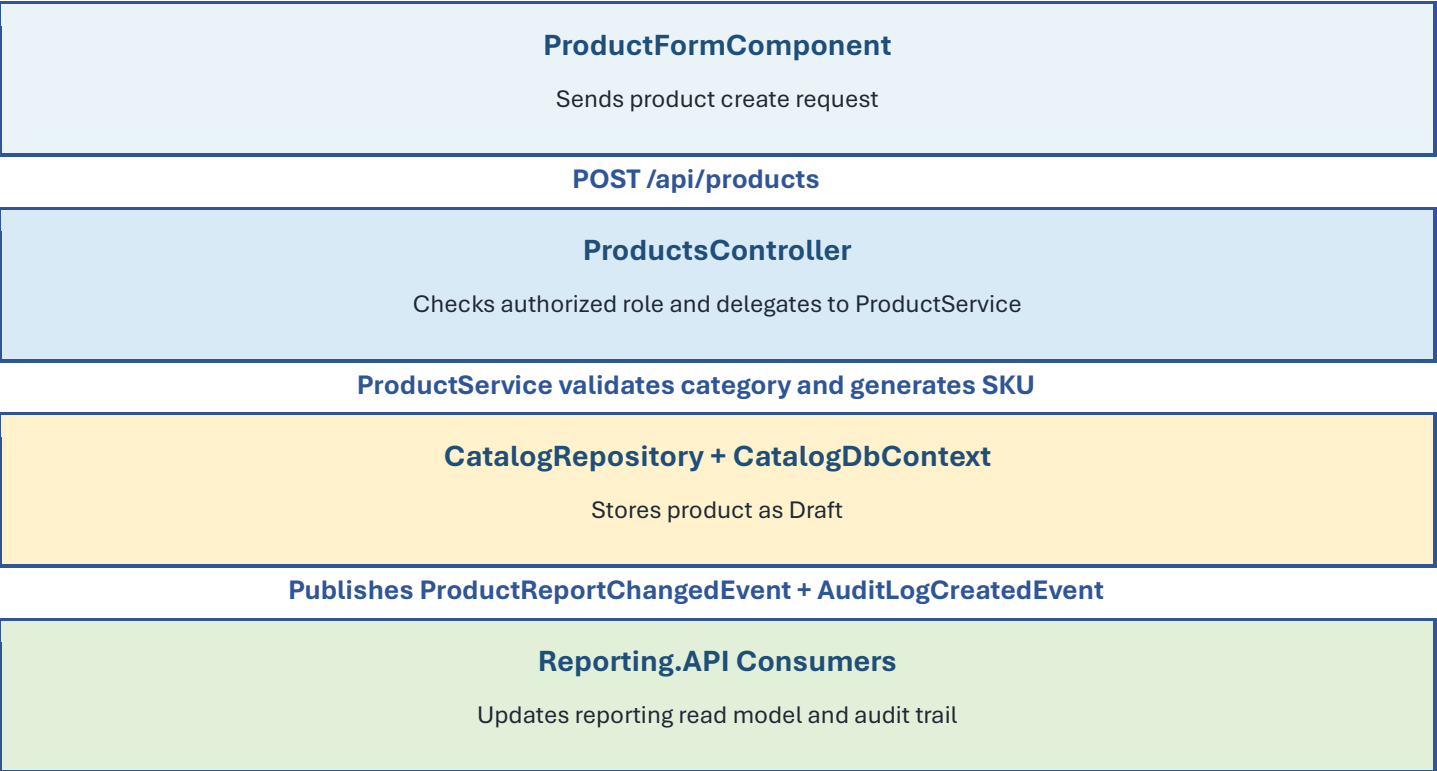
/api/workflow/products/{id}/submit	POST submit	WorkflowController	SubmitForReviewAsync.
/api/workflow/products/{id}/status	GET/PUT status	WorkflowController	GetApprovalStatusAsync, UpdateStatusAsync.
/api/reports	GET dashboard/products/export/snapshots	ReportsController	Reporting query and export methods.
/api/audit	GET audit logs	AuditController	Audit retrieval methods.

9. Low-Level Request Flows

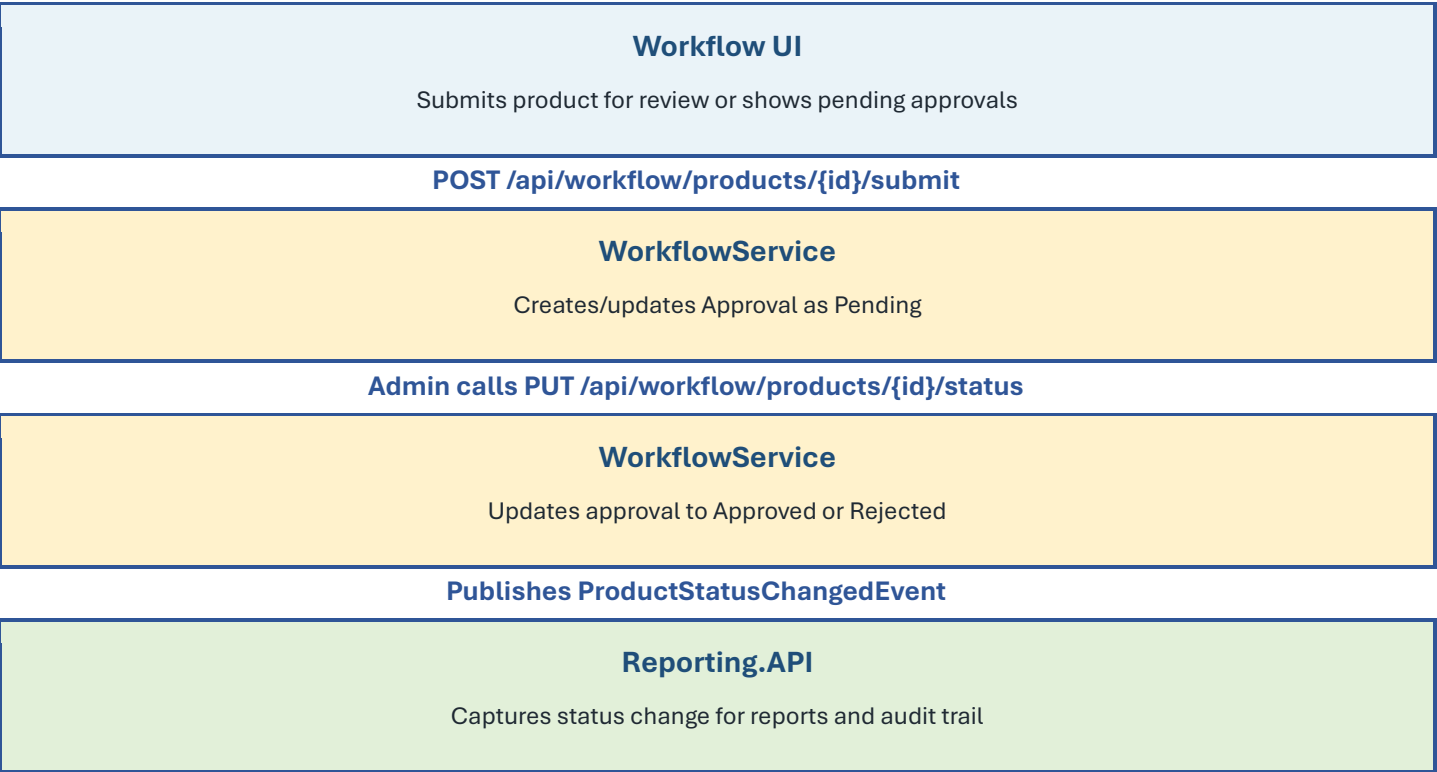
Login Flow



Product Creation Flow



Approval Flow



10. Event Contract Mapping

Event Contract	Important Fields	LLD Usage
AuditLogCreatedEvent	EntityName, EntityId, Action, ByUserId, OldValues, NewValues	Published when important user/product/workflow actions must be audit logged.
ProductReportChangedEvent	ProductId, ProductName, SKU, Status, CategoryName, IsDeleted	Published by Catalog.API to maintain product reporting data.
ProductStatusChangedEvent	ProductId, OldStatus, NewStatus, ChangedByUserId	Published by Workflow.API after admin approval status changes.
UserCountChangedEvent	Delta	Published by Identity.API when users are registered or deleted.

11. NUnit Low-Level Test Design

Test Type	Implementation Style	Purpose
Service Unit Tests	NUnit [TestFixture] and [Test] with Moq dependencies	Validate business rules without real infrastructure.
Repository Integration Tests	NUnit with EF Core InMemory database	Validate repository save/query behavior.
Controller Integration Tests	NUnit with WebApplicationFactory	Validate API endpoint behavior through in-memory test host.
Coverage Support	coverlet.collector in major test projects	Supports test coverage reporting for sprint evaluation.
Project	LLD Test Focus	
Catalog.API.Tests	Product, category, media asset, variant service/repository/controller behavior.	
Identity.API.Tests	Auth, user service, user repository, auth repository and controller behavior.	
Workflow.API.Tests	Workflow endpoint behavior for pricing, inventory and approval operations.	
Reporting.API.Tests	Reports controller and reporting service behavior.	